

P-V-Triggered Recalibration of Pressure-Only Soft-Actuator Proprioception: A Simulation Study

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Status: Unreviewed v1.4 correction candidate, updated 2026-09-29; not released or submitted. This revision retains the methods correction and the archived results in §§4.1–4.5. Section 4.6 incorporates the subsequently completed observability studies and their corrections from the repository's canonical results; this manuscript update does not rerun them. **This is a simulation-only modeling study.** No physical experiments are reported; every result is synthetic and is labeled as such. Modeling assumptions are stated as choices, not as calibrated predictions for a physical actuator. The natural next step is a physical bench campaign; the authors welcome hardware-validation collaboration.

Abstract

Soft pneumatic grippers face two coupled deployment problems: silicone chambers fail by cyclic fatigue, and recovering gripper pose without added sensors relies on the pressure signals already in the control loop. We study, in simulation, whether an idealized model-derived pressure-volume (P-V) hysteresis loop can serve as a *candidate recalibration signal* of fatigue-driven proprioception degradation in the practical shared-manifold topology, where chambers share one supply to cut valve count. We build a transparent, analytically invertible pipeline: a viscoelastic (standard-linear-solid) pneumatic plant producing rate-dependent P-V loops; a canonical fatigue model (Mullins softening with partial recovery, irreversible compliance drift, late leak growth); fatigue-coupled piecewise-constant-curvature (PCC) tip kinematics; a seeded sensor model; and a shared-manifold-versus-isolated-supply network simulator in which inter-chamber cross-talk lives in the actuation-band dynamics. From 2,000 labeled traces (20 actuators $\times$ 5 life stages $\times$ 2 supply topologies $\times$ 2 contact states $\times$ 5 repetitions), split by **actuator identity**, we evaluate static and dynamic pressure-only correctors and four recalibration policies (never, cycle-count clock, P-V-triggered, always-on).

We report four findings, including two negative/audit results. **(1)** A predicted advantage of a dynamic (lagged-input) corrector over a static ridge map under the shared manifold **does not materialize**: the cross-talk is real in the network dynamics but second-order, and the dynamic corrector improves pose error by $\approx 0\%$ at the assumed parameterization. **(2)** The dominant pressure-only proprioception error is instead the fatigue **compliance-scale drift**, which is topology-independent and causes a young-calibrated estimator's curvature error to grow by roughly two orders of magnitude over life. **(3)** The idealized P-V loop area is correlated with that drift within the shared generator (Pearson $r = 0.885$; 95% CI [0.853, 0.950] by an exhaustive actuator-cluster bootstrap on six held-out actuators, widening to [0.576, 0.973] under delete-one-actuator resampling), but a lead-time audit found **no positive temporal lead** under that deployed threshold on the 5-stage grid: the trigger crossing occurred at life 0.83, while the fixed-calibration budget crossing occurred earlier (median 0.71, range 0.56-0.71), giving median lead = -0.123 normalized life (6/6 held-out actuators nonpositive). A descriptive frontier shows why: lowering sensitivity to $\tau=0.01$ buys positive lead for all held-out actuators (median +0.346 normalized life) while still using 40% fewer recalibrations than always-on. **(4)** The deployed P-V-health-triggered recalibration policy meets a train-derived budget on macro-averaged stage pose RMSE at **60% fewer recalibrations** than an always-on policy (2 vs 5 per actuator), with the trigger threshold selected on training actuators only. A sensing-free cycle-count schedule tuned by the same rule fails that budget on held-out actuators (0.21 mm vs the 0.159 mm budget) at a similar recalibration count — the

idealized model-derived P-V signal performs better than that cycle schedule within this generator, not across real devices. **Scope caveat (stated up front):** all 20 actuators are drawn from a single generative model, so "generalization to unseen actuators" tests robustness to that generator, not real device-to-device variation; and absolute pose errors remain sub-millimeter throughout, so the operational value of triggered recalibration grows with tighter accuracy requirements.

Subsequent simulations with dispersed degradation laws narrow this interpretation: normalized-life estimation fails the declared unseen-unit transfer criterion, and corrected profile likelihoods do not establish post-onset life precision. These later studies examine a different endpoint from pose recalibration; they limit any interpretation of the original policy result as a transferable life sensor.

1. Introduction

Soft pneumatic grippers are attractive for safe, compliant manipulation, but two monitoring problems limit their deployment. **Fatigue:** silicone actuators fail by cyclic fatigue, and current practice detects failure only after rupture or via before/after diagnostics [Mosadegh 2014; Wong 2026]. **Proprioception:** recovering pose without added sensors uses the pressure signals already present in the control loop [L. Wang 2020]. Both problems are usually studied in isolation and, more importantly, under an *independent-chamber* assumption.

The most practical multi-finger topology shares a pneumatic manifold across chambers to reduce valve count and onboard mass [McDonald 2021; Moran 2024]. On a shared manifold, driving one chamber's valve perturbs the shared supply node and therefore its neighbors — inter-chamber **cross-talk**. This paper asks whether, in that topology, fatigue and proprioception are *physically coupled*: as a chamber fatigues its compliance rises, which (a) changes the P-V loop shape and (b) could change the cross-talk, so that a single pressure stream might monitor both health and pose.

We test this coupling in a deliberately transparent simulation, before any hardware. The modeling choices are made for analytic checkability (noise-free pose is recoverable to machine precision; isolated supplies give numerically zero cross-talk), and every estimator is validated against its own ground truth. The contribution of this paper is therefore the **pipeline, the design outputs, and a pre-specified modeling result framed as a prediction** (frozen in docs/result_spine.md, commit d856455, before result write-up) — not an experimental claim. The pre-specified temporal-lead expectation was audited explicitly and downgraded when the deployed threshold failed to show positive lead, which is the intended role of the frozen spine.

The later observability program asks whether the same interpretation survives variation in the degradation law itself. We report its completed results separately in §4.6 so that the original pose-recalibration experiment and the subsequent normalized-life estimation task retain their distinct endpoints, cohorts and decision rules.

2. Related work and prior-art boundary

P-V hysteresis and fatigue. P-V hysteresis is *already* used as a before/after fatigue diagnostic: Mosadegh et al. [2014] assess PneuNet durability across $>10^6$ cycles by comparing P-V curves, and US Patent 10,639,801B2 [Mosadegh, Shepherd, Whitesides 2020] documents the same before/after protocol. Libby et al. [2023] show the loop shifts with fatigue, and report FEM-agreement degradation (96%→80%). Hysteresis-loop area as a damage proxy has strong cross-domain precedent — metals [Haghshenas 2021], flight-control health indicators [Guo 2021], SHM/acoustic-emission RUL [Galanopoulos 2023] — grounded in early-warning theory [Scheffer 2009] and the prognostics-pipeline / health-indicator-quality criteria of Lei et al. [2018]. **We therefore do not claim first use of P-V for fatigue.** The evaluated contribution is a five-stage simulated recalibration-policy comparison with train-selected thresholds and a negative shared-manifold corrector result. Study 3 does not establish noisy-probe robustness or separation of drift from varying rest-history recovery; these require new interventions.

Pressure-only proprioception. Multi-chamber pressure-only sensing exists [L. Wang 2020; L. Wang 2023; J. Wang 2025; Joshi & Paik 2023; Zou 2024; Preechayasomboon & Rombokas 2021], but assumes independent or clean per-chamber supply. The nearest collision is Lindenroth et al. [2023], which senses contact force on *coupled parallel* chambers but never frames inter-chamber cross-talk as a proprioception confound or links it to fatigue.

Shared-manifold modeling. Lumped resistance-compliance models of pneumatic circuits [Stanley 2021] provide the tool to model cross-talk as a function of chamber compliance; shared single-source architectures are common in practice [Moran 2024; McDonald 2021].

Correctors and recalibration. ARX/echo-state correctors and physical reservoir computing are established for pneumatic hysteresis [Jaeger & Haas 2004; Youssef 2022; Shen 2025]. Online adaptation to soft-sensor drift exists but is *always-on* [Kushawaha 2025; Thuruthel 2022], or masks anomalous sensors [Sugiyama 2025]; a *fatigue-health-triggered* recalibration is the contrast we evaluate.

Ground truth and evaluation. We follow the soft-proprioception evaluation conventions of Zhang et al. [2023] and report pose RMSE (contact state is generated and stored in the dataset but is not an estimation target in this study — the correctors regress curvature/pose only, so no contact F1 is reported); PCC [Webster & Jones 2010] is treated as a modeled output, never as measured truth.

3. Modeling and methods

The pipeline is implemented in Python (`sim/`, `pipeline/`, `scripts/`) with unit-tested gates at each stage. All randomness is seeded; the dataset is reproducible from a fixed seed and its integrity is pinned by a manifest SHA-256.

3.1 Viscoelastic plant and P-V loops (Phase A)

Chamber walls are modeled as a standard linear solid (Zener) in the pressure-volume analog: a spring k_1 in parallel with a Maxwell arm (k_2 , relaxation time τ). This yields rate-dependent P-V hysteresis whose dissipation per cycle has the closed form $E_{\text{loss}}(\omega) = \pi k_2 A^2 (\omega\tau)/(1+(\omega\tau)^2)$, used as a ground-truth check on the numerical loop area, and reduces to the linear cross-talk model in the quasi-static limit. (*Validated: Phase A PASS.*)

3.2 Fatigue model (Phase B)

A canonical, deterministic life law injects: Mullins softening with a 30% permanent floor and 24-h recovery time constant; irreversible compliance drift; curvature-acceleration onset at 70% life; and late leak growth (to 20× conductance) observed by a separate closed-valve pressure-decay probe. Degradation enters the plant as a rising compliance multiplier (softer k_1) and loss multiplier (larger k_2). (*Validated: Phase B PASS, synthetic consistency.*)

3.3 P-V health features (Phase C)

Causal P-V features, Mullins-recovery normalization, health-index metrics, and matched false-alarm detectors were validated across a 144-condition identifiability grid; a matched segmented model recovers the quadratic onset and fails on logistic-onset variants — the intended inverse-crime/generalization control. (*Validated: Phase C PASS, synthetic.*) Phase C metrics are applied, not re-validated, downstream.

3.4 Fatigue-coupled PCC kinematics (Phase D)

Realized chamber pressure maps to bending curvature through the fatigue compliance multiplier, $\kappa = \kappa_{\text{gain}} \cdot C(\text{life}) \cdot \max(P - P_0, 0)$, so fatigue *amplifies* curvature at fixed pressure. Curvature maps to tip SE(3) by the robot-independent PCC convention [Webster & Jones 2010]. The map is analytically invertible: forward↔inverse round-trips to machine precision, and a noise-free pose is recoverable exactly (gate test).

3.5 Sensor model

A seeded measurement model applies Gaussian noise, quantization, decimation (sampling), and a contact observation (binary flag + synthetic normal force from tip penetration). Independent sub-streams per channel make the corruption reproducible and channel-order-independent. These corruptions apply to dataset observations, not the separate idealized Study 3 health signal described in §3.9.

3.6 Shared-manifold vs isolated network dynamics

A time-domain integrator drives a focal chamber plus two neighbors. In the **shared** topology all chambers draw from one supply node; driving one valve pulls the shared node and perturbs the neighbors. In the **isolated** topology each chamber has its own regulated node and the state blocks are decoupled. Consistent with a lumped-RC pre-test (Gate 0), the cross-talk is *dynamic*: its DC gain is compliance-independent while its actuation-band gain drifts with fatigue. Excitations are multisine in the 1–5 Hz actuation band so the effect is observable. Gate tests confirm: isolated coupling is numerically zero, shared coupling is measurable, and the DC-independent / actuation-band-drift property holds.

3.7 Dataset

20 actuators × 5 life fractions {0.1,0.3,0.5,0.7,0.9} × 2 topologies × 2 contact states × 5 repetitions = **2,000 traces** (240 time samples each). Per-actuator geometry, wall stiffness, and rupture life are drawn deterministically. The split is by **actuator identity** (14 train / 6 held-out), never by trace, so generalization is measured on actuators never seen in training. Features stored per trace include the shared-manifold pressure, all chamber valve commands, the focal chamber pressure and volume, the true and measured tip positions, the true curvature, and contact.

3.8 Proprioception correctors (Phase E)

The target is focal bending curvature (and, through PCC, tip position) from pressure-only features [shared-manifold pressure, all chamber commands]. We compare a **static ridge** map (memoryless) against a **dynamic** corrector that is the identical ridge estimator on lagged inputs (an exogenous ARX/FIR form; output feedback is excluded because the pose is the estimation target, not an online measurement). The lag count is the only difference.

3.9 P-V health signal and recalibration policies (Phase F)

The health signal is an **idealized model-derived** P-V loop area, normalized to its young value. `scripts/run_study3.py` calls `pipeline/coupling.py::health_trajectory`, which computes analytic SLS loop area with zero rest input to `fatigue_state`. It does not use the noisy, quantized, decimated volume observations in the Phase D dataset. Thus probe-noise and variable-rest robustness are not tested here. Three policies, each starting from one calibration at the youngest life stage: **fixed** (never recalibrate), **always** (recalibrate every stage), and **triggered** (recalibrate when fractional loop-area growth since the last calibration exceeds τ). The threshold τ is selected on **training actuators only** against a stated accuracy budget, then applied unchanged to held-out actuators. We report estimation error **and** recalibration count together, so a "savings" cannot hide degraded accuracy.

3.10 Subsequent observability studies: a distinct evaluation

Studies A-C use a dispersed synthetic cohort to examine between-unit health-signal variation, local information about normalized life, and transfer of a life estimator, respectively. Their inputs, parameter provenance and imposed acceptance criteria are recorded in the parameter register; they are modeling choices rather than hardware-calibrated quantities. Rupture life is assigned by the generator, not inferred from a common measured damage threshold.

Study B's corrected profile analysis compares an aged snapshot alone with a stacked design that also observes a young baseline under shared nuisance parameters. The baseline's normalized-life coordinate is treated as known: a known cycle count alone would not supply that information without knowledge of rupture life. Study C instead schedules probes by cycle count and fits a ridge estimator to baseline-normalized probe features, their lagged values and the clock, with tuning confined to training units. The probe vector includes P-V loop and stiffness features as well as pressure-decay features; it is not a demonstrated pressure-only acquisition system. This distinction leaves the pressure-only *pose corrector* in §3.8 intact while making the health-probe sensing requirement explicit.

Study C's channel-muting analysis is post-hoc and retains the fitted weights. The subsequent dispersion sweep is a sensitivity audit within the same model family. Neither is an independent confirmation cohort or a new recalibration-policy evaluation.

4. Results

4.1 Validation gates

All structural gates pass: noise-free pose recovery and forward↔inverse curvature round-trips to machine precision; isolated-supply off-diagonal coupling at the solver-noise floor ($<10^{-6}$) while shared coupling is measurable ($>10^{-4}$); and the Gate-0 property that DC cross-talk is compliance-independent while actuation-band cross-talk drifts with fatigue. The full unit-test suite gates these checks.

4.2 Cross-talk is real but second-order (negative result, retained; Fig. 1)

The proposal predicted that shared-manifold cross-talk drift would be visible to a dynamic corrector and invisible to a static ridge, making the dynamic corrector win under the shared manifold. **This does not occur.** With all chamber commands available as features and per-actuator calibration, the dynamic (lagged-input) corrector improves curvature RMSE over the static ridge by $\approx 0\%$ **under either topology** (shared: 0.244 vs 0.244 1/m; isolated: 0.295 vs 0.295 1/m; pose ≈ 0.84 vs 0.84 mm and 1.02 vs 1.02 mm). The cross-talk is genuine in the network dynamics (§4.1) but is a small perturbation ($\approx 4\%$ of the focal curvature at the selected model parameters) that does not materially degrade pose estimation. We retain this as a negative result and did not tune network parameters to manufacture the predicted effect; §5 characterizes the envelope over which it holds by sweeping the supply softness.

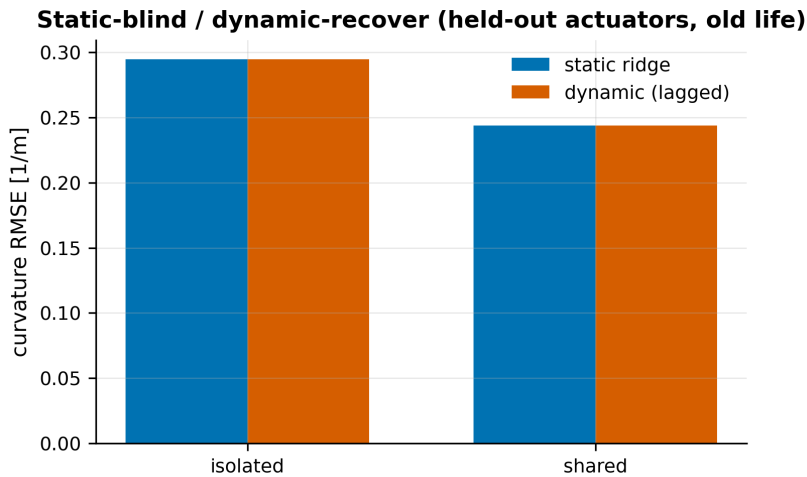


Figure 1. Static vs dynamic (lagged-input) corrector on held-out actuators at old life; the dynamic corrector yields $\sim 0\%$ improvement under either supply topology — the predicted shared-manifold cross-talk advantage does not appear.

4.3 Compliance drift dominates the proprioception error (Fig. 2)

A young-calibrated static estimator's curvature RMSE grows by roughly two orders of magnitude over life ($\approx 0.004 \rightarrow \approx 0.46\text{--}0.56$ 1/m at 0.9 life), and the shared and isolated curves are comparable — the difference does not consistently favor the shared-manifold-degrades-more hypothesis. The dominant pressure-only proprioception error is therefore the **topology-independent fatigue compliance-scale drift**, not cross-talk. This both explains §4.2 and motivates recalibration as the right intervention.

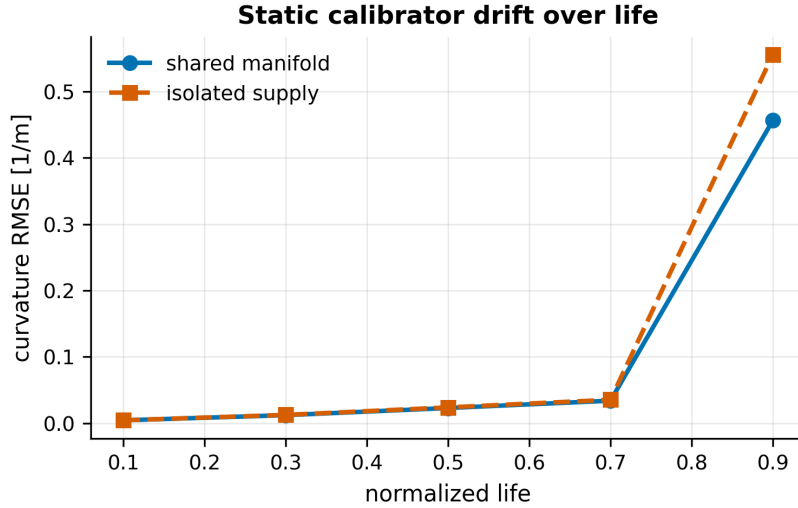


Figure 2. Young-calibrated static estimator curvature RMSE over normalized life (shared vs isolated supply); error grows ~100x by end of life and is comparable across topologies — the degradation is the topology-independent compliance drift, not cross-talk.

4.4 P-V loop area is a health indicator, not a positive-lead trigger on this grid (Fig. 3)

On held-out actuators, the idealized P-V loop-area fractional growth tracks the fixed-calibration pose error over life with **Pearson $r = 0.885$** . The 30 points are 6 actuators x 5 life stages, i.e. repeated measures on six clusters, so the interval is computed with the actuator as the resampling unit: an exhaustive nonparametric cluster bootstrap over all $6 \times 6 = 46,656$ actuator resamples gives **95% CI [0.853, 0.950]**, and a delete-one-actuator jackknife on Fisher z — the more conservative reading, with six effective observations — gives **[0.576, 0.973]**. Both exclude 0. For comparison, the point-level bootstrap used in v1 (2,000 resamples of the 30 points treated as independent) gives [0.835, 0.958]; it is *not* anti-conservative here, because the clusters are nearly homogeneous, but it overstates the effective sample size and is superseded. All intervals are reproduced by `scripts/run_study3_cluster_ci.py`. The same monotone structure appears within actuators: per-actuator r has median 0.957 and range [0.9570, 0.9574] across the six held-out actuators (5 stages each). That tight range is not independent evidence from six distinct health trajectories: in this generator, young-normalization makes the P-V health trajectory identical to numerical precision across actuator parameter draws. The per-actuator result is therefore a decoded generator property and part of the single-generator caveat, not a claim of real device-to-device health-signal diversity.

The temporal lead audit has three parts. First, at the deployed train-selected $\tau^*=0.05$ threshold, the trigger crossing occurs at life 0.83 for each held-out actuator, while the young/fixed-calibration error crosses the 0.159 mm budget earlier: median life 0.71, range 0.56-0.71. Linear interpolation between the five life stages gives median lead = -0.123 normalized life, range [-0.269, -0.115] (about -972 to -375 cycles across the held-out rupture lives); all 6/6 held-out actuators are nonpositive-lead cases. Second, a descriptive frontier shows that lead is purchasable, not free: $\tau=0.01$ triggers at median life 0.36 with median lead +0.346 normalized life (range +0.199 to +0.354), uses 3 recalibrations per actuator, and remains within budget at 0.029 mm; $\tau=0.02$ still meets budget at 0.037 mm with 3 recalibrations, but 1/6 held-out actuators has nonpositive lead; that case is actuator 17, the same early budget-crossing held-out actuator that dominates the deployed threshold's worst negative lead. At the most sensitive frontier point, $\tau=0.005$ fires at every life stage and is identical to always-on (5 recalibrations, 0.019 mm). Third, the deployed $\tau^*=0.05$ point fires late because the normalized signal's full dynamic range is only 6.5% over life, so τ^*

consumes about 77% of the available range. Thus the idealized loop area is a strong health correlate and useful recalibration trigger, but positive temporal lead is configuration-dependent and is not a title-level claim.

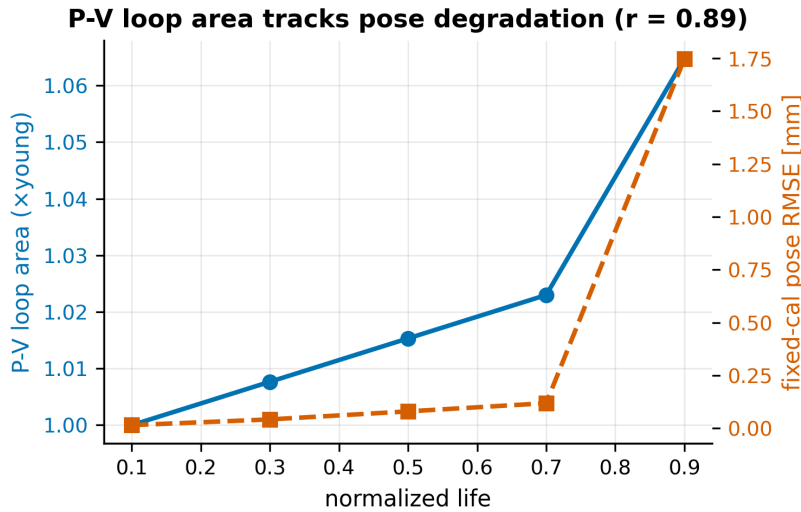


Figure 3. Idealized P-V loop-area growth (left axis) tracks fixed-calibration pose error (right axis) over life on held-out actuators; $r = 0.885$, 95% CI [0.853, 0.950] (actuator-cluster bootstrap). The deployed threshold does not provide positive temporal lead, but lower thresholds define a lead-vs-recalibration frontier.

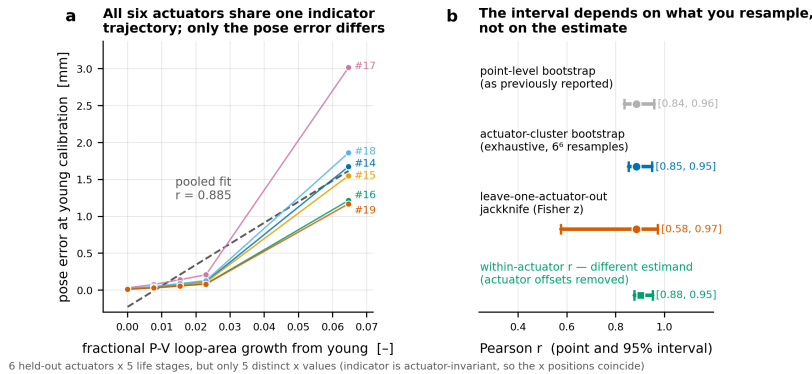


Figure 3b. (a) Per-actuator scatter behind the pooled correlation. Because young-normalization makes the health trajectory actuator-invariant (§4.4), the six actuators share the same five x positions and differ only in pose error; the pooled fit is shown dashed. (b) The same estimate $r = 0.885$ under three resampling units: point-level (as in v1), exhaustive actuator-cluster bootstrap, and delete-one-actuator jackknife on Fisher z . The interval width varies roughly fourfold with the choice of resampling unit while the estimate does not. The within-actuator correlation (actuator offsets removed) is a different estimand and is marked as such.

4.5 Recalibration trade-off (Fig. 4)

The endpoint is the macro-average of stage-level pose RMSE across five sampled life stages, followed by averaging across actuators (policy_metrics in scripts/run_study3.py). It is not a maximum error or a guarantee between stages. Against a train-derived 0.159 mm budget (selected on training actuators as halfway from the always-on error toward the never-recalibrate error), we compare four policies. Besides fixed and always-on, the P-V trigger must beat the obvious *sensing-free* competitor: a **cycle-count clock** that recalibrates every T actuation cycles. Both adaptive policies are tuned on training actuators by the same rule (fewest recalibrations within budget): $\tau^* = 0.05$ fractional loop-area growth for the trigger, $T^* = 2,700$ cycles for the clock (train errors 0.044 and 0.132 mm respectively — both within budget on train). On held-out actuators:

Policy	Macro-averaged stage pose RMSE (held-out)	Calibration events / actuator, including initial	Meets mean 0.159 mm budget?
fixed (never)	0.40 mm	1	no
cycle-count clock ($T^*=2,700$)	0.21 mm	1.5	no — fails on transfer
P-V-triggered ($\tau^*=0.05$)	0.06 mm	2	yes (2.7× margin)
always-on	0.02 mm	5	yes

Two comparisons matter. Against always-on, the trigger holds pose error near the always-on band at **60% fewer recalibrations** (2 vs 5 per actuator). This ratio is quoted without a confidence interval by design: every held-out actuator produces the same counts under both policies (2 and 5), a direct consequence of the actuator-invariant indicator (§4.4) crossing any threshold at the same life fraction. The saving is therefore a deterministic property of the 5-stage life grid at τ^* , not an estimate with between-actuator sampling variability, and an interval on it would be degenerate ([60.0%, 60.0%]). Only the sensing-free clock baseline varies across actuators (1.50 recalibrations, 95% cluster CI [1.17, 1.83]). Against the clock — the comparison that isolates what the *model-derived P-V signal* buys in this experiment — the clock, tuned within budget on training actuators, **violates the budget on held-out actuators** (0.21 mm > 0.159 mm) at a similar recalibration count, because a global cycle period misaligns with per-actuator degradation state whenever rupture life varies across devices (here drawn from 3,000–4,000 cycles), while the trigger is effectively a normalized-life sensor in this generator and meets the reported mean budget on its held-out identities. Absolute errors are sub-mm in all policies (see §5), so the demonstrated value is the error-vs-cost trade-off and its transfer to unseen actuators, not the absolute accuracy.

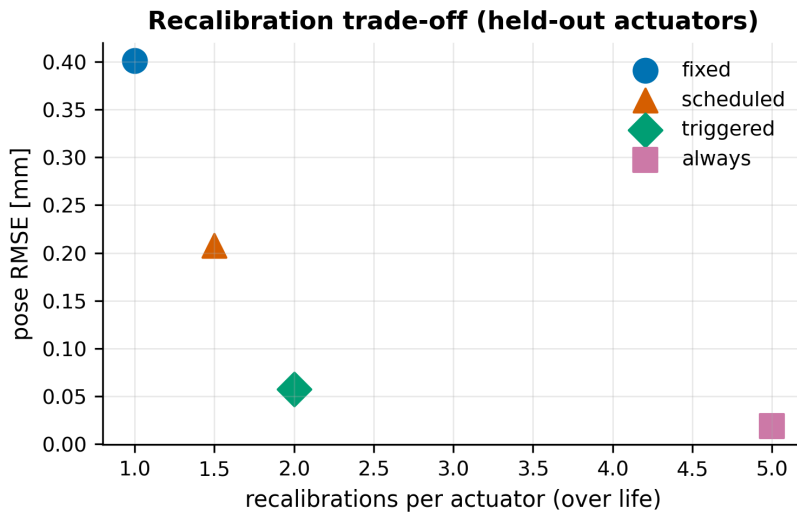


Figure 4. Recalibration trade-off on held-out actuators — pose error vs recalibrations per actuator for fixed, cycle-count clock, P-V-triggered, and always-on policies. The triggered policy meets the accuracy budget at 60% fewer recalibrations than always-on; the sensing-free clock, tuned within budget on training actuators by the same rule, fails the budget on transfer.

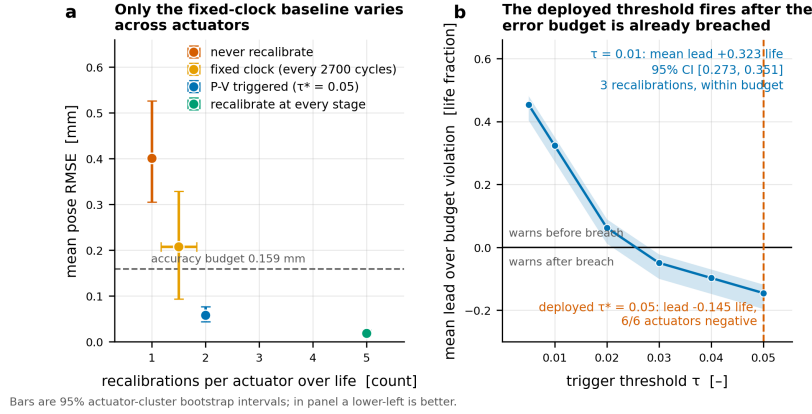


Figure 4b. (a) The same four policies with 95% actuator-cluster bootstrap intervals on both axes. Only the fixed-clock baseline has non-zero between-actuator spread in recalibration count; the triggered and always-on counts are identical on every held-out actuator, which is why the 60% saving carries no interval (\$4.5). (b) Lead-time frontier with cluster intervals. Both axes are actuator **means**, so the lead plotted here is the mean over held-out actuators, not the median quoted in \$4.4: at $\tau = 0.01$ the mean lead is +0.323 normalized life (95% cluster CI [0.273, 0.351]) against a median of +0.346, the difference coming from actuator 17 at +0.199. The deployed $\tau^* = 0.05$ sits below zero lead for 6/6 actuators; $\tau = 0.01$ keeps 3 recalibrations per actuator and stays within the accuracy budget at 0.029 mm.

4.6 Completed observability follow-ups and limits on transfer

Study A: variation depends on the degradation law. Dispersion of degradation-law parameters breaks the young-normalized trajectory invariance seen in the original generator. The original rule nevertheless returned A-FAIL because its trigger-timing requirement did not pass. An owner amendment removed that requirement from the cohort-admissibility gate, yielding A-PASS under the amended rule. This is a changed criterion, not an improved result; both verdicts and the ablations remain in the Study A result. It establishes a nondegenerate synthetic cohort for a transfer test, not successful transfer.

Study B: post-onset precision is not established. The corrected analysis uses the stated stacked observation design, scale-stable covariance whitening and log-space optimization of positive nuisance parameters. Undefined decay features are treated as infeasible rather than passed to the optimizer. In the canonical post-onset case, the resulting profile has a broad, nearly flat region whose terminations coincide with nuisance bounds. Widening the nuisance box removes the upper threshold crossing. The stored identifiable label therefore describes crossings under a particular box, not demonstrated structural uniqueness or a stable precision estimate. Both previously reported resolution figures are withdrawn; the corrected profile result is the canonical record of the profiles, bound sensitivity and null resolution field.

The known-coordinate young baseline helps localization within that specified synthetic design but does not establish a deployable baseline procedure. The pre-onset profiles retain unresolved optimizer statuses. The canonical-unit profile uses one noise realization, so its point-estimate error is not an estimate of bias. The likelihood threshold has not been coverage-calibrated here, and the profile endpoints must not be interpreted as a validated confidence interval. Strong joint collinearity remains a local diagnostic; numerically singular subsets have a status rather than a reportable index magnitude. The retained Fisher-map result does not override these limits.

Study C: the life estimator fails the declared transfer test. The estimator does not satisfy the joint requirements on held-out life-estimation error and improvement over the clock baseline. The verdict remains C-FAIL; per-unit errors, uncertainty summaries and the adversarial search are in the Study C result. The post-hoc failure analysis associates error more strongly with mismatch to the

training cohort's rupture-life clock than with post-onset probe coverage. Muting the clock worsens aggregate performance while helping the worst unit. Coverage and clock mismatch are correlated by the fixed-cycle schedule, so this diagnostic cannot isolate their causal effects or show that denser probing would solve transfer.

Dispersion sensitivity: no tested setting clears the joint rule. In the dispersion audit, lower lifetime dispersion improves absolute life-estimation accuracy but also makes the clock baseline harder to beat. At the original stress-case dispersion, relative improvement over the clock is more common but absolute accuracy is insufficient. The failure persists across the tested settings for different reasons; this does not establish impossibility at untested settings or for other estimators. The audit also shows that units reach their assigned rupture at different degradation states, an imposed modeling choice rather than evidence of a shared physical failure threshold.

These follow-ups restrict the interpretation of §§4.4–4.5 without replacing their archived pose-policy results. The proposed Study C2 has a preregistration and runner but no executed study result; no C2 outcome is included in this manuscript.

5. Discussion and limitations

- **Simulation only.** Every result is synthetic. The pipeline and the modeling result (framed as a prediction) are the contributions; no experimental validation is claimed.

- **Limited generator variation.** The original cohort shares one degradation law; the later studies disperse selected parameters within the same model family. Neither tests real device-to-device spread or independent failure mechanisms. The later C-FAIL result limits the interpretation of the original within-generator policy advantage.

- **Correlation is structurally favored, not demonstrated diagnostic robustness.**

Both loop area and pose drift share the fatigue generator. Study 3's analytic health trajectory has no injected probe noise, quantization or decimation, and uses zero rest input. Those mechanisms cannot explain its correlation interval. Pose-channel noise and estimator/PCC behavior affect the other variable, but are not a substitute for health-signal perturbations. Young-normalization makes the health trajectory effectively identical across actuators. The intervals describe resampling uncertainty among six synthetic identities, not robustness to alternative failure modes or real devices. Leak-dominated, sensor-biased and local-damage interventions must be tested separately.

- **The headline cross-talk hypothesis was not supported.** Cross-talk is monotone in compliance

(Gate 0) but second-order for pose at the tested parameters. The proposal anticipated this "benign coupling" outcome as a publishable floor; the deliverable is the health-indicator + recalibration result via the compliance-drift pathway.

- **Parameter-sensitivity of cross-talk (envelope, Fig. 5).** To make the "second-order" claim a characterized regime rather than a single-point assertion, we swept the network parameters that set the coupling magnitude and measured the neighbor/driven response ratio at the actuation band (probe_coupling, scripts/run_study4.py). The coupling rises monotonically with supply softness: at the default parameters it is **6.3%** (a response ratio, not the $\approx 4\%$ curvature share quoted in §4.2), and it crosses the **10%** "starts to matter" line only when the supply resistance R_s is $\approx 1.7\times$ softer and the **20%** line at $\approx 4.3\times$ softer. The manifold compliance C_m is a far weaker knob — a larger buffer reduces coupling, and across a $128\times$ span ($\times 0.25$ to $\times 32$) it stays in the **5.8–6.3%** band and never reaches 10%. So the negative result is not knife-edge: cross-talk remains second-order across the tested model configurations and would only become first-order for pose under a deliberately under-provisioned (several-fold throttled) supply. A confirmatory corrector check at the softer

settings is consistent — the dynamic (lagged) corrector still does not beat the static ridge at these small-sample sizes, though its penalty shrinks as coupling grows.

- **Sub-millimeter absolute errors.** Even never-recalibrating yields ≈ 0.40 mm pose RMSE at this sensor-noise level, so the operational case for triggered recalibration strengthens with tighter pose requirements, softer actuators (larger per-stage drift), or higher noise.

- **Modeling choices.** The fatigue law, PCC single-segment kinematics, and lumped network are transparent assumptions chosen for analytic checkability, not calibrated fits. The parameter register distinguishes imposed criteria, selected model values and literature-anchored assumptions from measured inputs. Simulation acceptance is internal consistency, not hardware validation.

- **Hardware status.** The repository now contains a known-volume reference-chamber CAD design. It is an instrumentation reference, not a soft-actuator specimen or a completed physical validation campaign. No specimen measurement supports the manuscript's results.

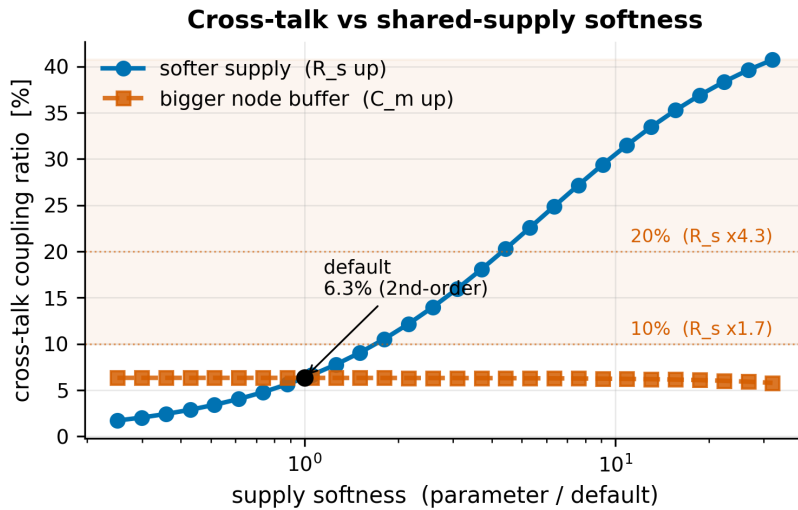


Figure 5. Cross-talk coupling ratio (neighbor/driven response at the actuation band) vs shared-supply softness, as a multiple of each default parameter. Coupling rises monotonically with supply resistance R_s and is $\approx 6.3\%$ at the default operating point (marked); it reaches the 10% and 20% "first-order" lines only at $R_s \approx 1.7\times$ and $\approx 4.3\times$ softer. Manifold compliance C_m is a weak knob (stays 5.8–6.3% across $\times 0.25$ – $\times 32$). The negative cross-talk result holds across the tested model configurations.

6. Conclusion

Within one simulation generator, idealized P-V loop area is a candidate recalibration signal correlated ($r \approx 0.89$) with fatigue-driven pressure-only proprioception drift in shared-manifold soft grippers, but the deployed threshold does not show positive temporal lead on the 5-stage grid. A P-V-triggered policy meets the train-derived macro-averaged stage-RMSE budget with fewer counted calibration events (including initialization) than always-on adaptation, on held-out identities from the same generator — where a sensing-free cycle-count schedule tuned by the same rule does not. The shared-manifold cross-talk, while real, is second-order for pose at the tested parameters — a retained negative result. The subsequent dispersed-cohort studies do not establish transferable life estimation, and the corrected profile analysis supports no post-onset precision claim. The defensible contribution is thus a reproducible synthetic policy comparison with explicit negative results and limits, rather than a validated life sensor. Physical measurements of a defined specimen and health-probe acquisition chain remain necessary before deployment claims.

7. Reproducibility

All code, the dataset generator, and the analysis scripts are in the public repository (500ft/soft-actuator-recalibration); the archived numerical results in §§4.1–4.5 are pinned by the annotated tag preprint-v1.3. This version-controlled correction candidate and the later results in §4.6 are not part of that tag. The candidate source hash for review is recorded in publication-readiness.json; author approval remains pending. The dataset is regenerable from a fixed seed (python -m scripts.phaseD_dataset; integrity pinned by the manifest SHA-256); the studies are python -m scripts.run_study2 (correctors), python -m scripts.run_study3 (health indicator + recalibration), and python -m scripts.run_study4 (cross-talk parameter-sensitivity envelope). Frozen result JSON and figures are committed under data/sim/phaseD/; the pre-specified claim and target figures are in docs/result_spine.md (commit d856455, before result write-up). The full unit-test suite and manuscript-number check gate the pipeline in CI. The checker compares selected original manuscript numbers with committed outputs; it does not certify every scientific interpretation.

The later canonical result files are linked at each claim in §4.6. Their generators are scripts/run_studyA.py, scripts/run_studyB.py, scripts/run_studyB_structural.py, scripts/run_studyC.py, scripts/analyze_studyC_failure.py and scripts/run_dispersion_audit.py. The Study B correction record identifies the superseded analysis and the numerical repairs. These committed artifacts, rather than a fresh regeneration, supply this review candidate. Author review, a separately versioned reviewed PDF and a deposit decision remain required before publication.

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Citations follow the annotated literature base (docs/A01_A04_Literature_Review.md), where full DOIs are listed. Selected works cited here:

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